
Course Name: Machine Learning for Earth System Sciences

Week 01 - Assignment 01 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: $10 \times 1 = 10$

QUESTION 1

A meteorologist observes that if one weather station records heavy rainfall, nearby weather stations are also likely to record similar rainfall amounts. This phenomenon is best explained by:

- a) Temporal autocorrelation
- b) Spatial autocorrelation
- c) Random sampling
- d) Seasonal variability

Answer: (b)

Detailed Solution:

Rainfall at nearby locations tends to be similar because atmospheric systems (such as clouds and monsoon bands) affect neighbouring areas together. This follows the First Law of Geography, where nearby locations are more strongly related than distant ones.

QUESTION 2

For an autoregressive process of order-1 if $X_{t,0}$ represents temperature at time t_0 and has value 20°C at a particular location. Find out the temperature $X_{t,2}$ for additive white noise of 0.5 and regression coefficient $a=2$.

- a) 12°C
- b) 40.5°C
- c) 81.5°C
- d) 81.25°C

Answer: (c)

Detailed Solution:

An autoregressive process of order-1 is represented as:

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$X_{t,i+1} = a X_{t,i} + b$, where a is the regression coefficient and b is the white noise

$$\Rightarrow X_{t,1} = a X_{t,0} + b$$

Given $X_{t,0} = 20^{\circ}\text{C}$, $a=2$ and $b=0.5$

$$\Rightarrow X_{t,1} = 2 \cdot 20 + 0.5 = 40.5^{\circ}\text{C}$$

Now

$$X_{t,2} = a X_{t,1} + b$$

$$\Rightarrow X_{t,2} = 2 \cdot (40.5) + 0.5 = 81.5^{\circ}\text{C}$$

QUESTION 3

Which of the following is true regarding stationary processes

- a) White noise is the simplest example of a mean stationary process
- b) For a covariance stationary process, the first two moments (mean and variance) doesn't change over time
- c) For a covariance stationary process, the covariance of the time series with lagged values of itself is constant
- d) All the statements are TRUE.

Answer: (d)

Detailed Solution:

All statements are true regarding stationary time series.

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QUESTION 4

A positive sea surface temperature anomaly over the tropical ocean generally indicates:

- a) The sea surface is cooler than its climatological average.
- b) The sea surface is warmer than its climatological average
- c) The ocean has become shallower.
- d) Ocean salinity has increased

Answer: (b)

Detailed Solution:

SST anomalies compare the observed sea surface temperature with its long-term average. A positive SST anomaly means the observed sea surface temperature is above the long-term average for that location and time.

QUESTION 5

Which of the following statement(s) is/are true

- I. Gaussian Process Regression is a non-parametric technique.
- II. Gaussian Process Regression do not attempt to identify best fit models of the data, instead they compute a posterior distribution over models.

- a) I only
- b) II only
- c) Both I and II
- d) None of these

Answer: (c)

Detailed Solution:

Bayesian algorithms do not attempt to identify “best-fit” models of the data (or similarly, make “best guess” predictions for new test inputs). Instead, they compute a posterior distribution over models (or similarly, compute posterior predictive

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distributions for new test inputs). These distributions provide a useful way to quantify our uncertainty in model estimates.

QUESTION 6

Which of the following is NOT a component of a typical geostatistical model used for climate variables such as rainfall or temperature?

- a) Trend (mean) component
- b) Spatially correlated process
- c) Random noise component
- d) Activation function

Answer: (d)

Detailed Solution:

Activation functions are components of neural networks, not classical geostatistical models.

QUESTION 7

The primary role of a covariance (kernel) function in Gaussian Process Regression is to:

- a) Estimate the regression coefficients.
- b) Measure the similarity between input points.
- c) Normalize the input data.
- d) Compute the prediction error.

Answer: (b)

Detailed Solution:

The kernel quantifies how strongly two input locations are related, thereby determining the covariance between their function values.

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QUESTION 8

Consider 5 points with respective coordinates: A(5,5), B(5,6), C(6,7), D(6,6) and E(9,10). Identify the noise point among these points using DBSCAN clustering if $Eps=2$ and $Minpts=3$

- a) B(5,6)
- b) E(9,10)
- c) A(5,5)
- d) C(6,7)

Answer: (b)

Detailed Solution:

On applying DBSCAN on the given data points we can identify that E(9,10) is the outlier point (noise point).

Step 1: Compute distances

Step 2: Find ϵ -neighbourhoods

Since E is not within Eps of any core point, it is not a border point either.

Therefore, E(9,10) is classified as a noise (outlier) point.

QUESTION 9

A model predicts daily air temperature using historical temperature, humidity, and solar radiation. Which of the following is/are exogenous variables?

- a) Historical temperature only
- b) Humidity and solar radiation
- c) Future temperature values
- d) Prediction errors

Answer: (b)

Detailed Solution:

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Exogenous variables are external inputs used to help predict the target variable. The variable historical temperature is part of the autoregressive signal, not exogenous.

QUESTION 10

A weather forecasting lab wants to estimate rainfall at unobserved locations while also identifying regions where predictions are less reliable. Which machine learning technique is most suitable for this objective?

- a) Linear Regression
- b) Gaussian Process Regression
- c) K-Means Clustering
- d) Principal Component Analysis

Answer: (b)

Detailed Solution:

Gaussian Process Regression (GPR) models spatial correlation. By leveraging a spatial covariance function (kernel), it can smoothly interpolate rainfall values across a continuous geographic space. A core feature of GPR is uncertainty quantification.